

EXCEL ACADEMY

LAWS OF MOTION

Class 11 Physics - Chapter 4

Detailed Notes

Simple language • Correct LaTeX equations • Exam-focused numericals

Inertia, Newton's laws, momentum, impulse, conservation of momentum, equilibrium, friction, circular motion and free-body diagrams

How to use these notes

- First understand the force-motion logic. Then learn each formula together with its conditions.
- Draw a free-body diagram before writing equations in a numerical problem.
- Revise the misconceptions and problem-solving checklist before the examination.

1. Chapter Overview

- **Kinematics** describes motion through position, velocity and acceleration. This chapter explains **why** motion changes.
- A force may start motion, stop motion, change speed or change direction.
- **Contact forces** include normal reaction, friction, tension, air resistance and viscous drag.
- **Non-contact forces** include gravitational force and, where relevant, electric and magnetic forces.
- Galileo and Newton showed that a force is needed to **change velocity**, not to maintain a constant velocity when no resistance acts.

Central idea

- Zero net external force means zero acceleration, not necessarily zero velocity.
- A net external force changes momentum. Interactions always involve equal and opposite forces on different bodies.

2. Aristotle's Fallacy

- Aristotle believed that a continuous external force was needed to keep a body moving.
- This seems natural because a toy car stops when we stop pulling it.
- The hidden cause is **friction**. The car stops because the floor exerts a force opposite to its motion.
- While the car moves with constant velocity, the pull balances friction:

$$\vec{F}_{\text{net}} = \vec{F}_{\text{pull}} + \vec{f} = \vec{0} \quad \Rightarrow \quad \vec{a} = \vec{0}.$$

- If friction were absent, no continuous force would be required to maintain uniform straight-line motion.
- Aristotle converted everyday experience, where friction is present, into an incorrect general law.

Corrected statement

An external force is not required to maintain constant velocity. A net external force is required to change velocity or to balance resistance.

3. Galileo and the Law of Inertia

3.1 Single inclined-plane reasoning

- A body moving down an inclined plane accelerates.
- A body moving up an inclined plane slows down.
- A horizontal plane is the intermediate case. With no friction, a body on it neither speeds up nor slows down.

3.2 Double inclined-plane experiment

- A ball released on one smooth slope rolls down and climbs the opposite slope.
- Ideally, it reaches the same vertical height from which it was released.
- If the second slope is made gentler, the ball travels farther to reach the same height.
- If the second plane becomes horizontal, the ball would need to travel an infinite distance to regain its height. Thus, without friction, it would continue moving indefinitely.

3.3 Meaning of inertia

- Rest and uniform straight-line motion are equivalent because both have zero acceleration.
- **Inertia** is the tendency of a body to resist a change in its state of rest or uniform motion.
- A body changes that state only when acted upon by a net external force.

3.4 Ideas on motion in ancient Indian science

- Ancient Indian thinkers discussed continuous pressure (*nodan*), impact (*abhighat*), transmitted force and the persistent tendency called *sanskara*.
- *Vega*, the tendency to move in a straight line, comes close to the modern idea of inertia. Contact and the atmosphere were understood to oppose it.
- Bhaskara described instantaneous motion (*tatkaliki gati*), anticipating the modern idea of instantaneous velocity.
- A water current, which carries particles, was distinguished from a wave, which transmits vibrations.

4. Newton's First Law of Motion

Statement

Every body continues in its state of rest or uniform motion in a straight line unless compelled by a net external force to change that state.

$$\vec{F}_{\text{net}} = \vec{0} \implies \vec{a} = \vec{0}$$

- If a body is at rest and the net force is zero, it remains at rest.
- If a body is moving and the net force is zero, it continues with constant velocity in a straight line.
- On Earth, zero net force usually means that several forces cancel; it does not mean that no forces act.
- For a book at rest on a horizontal table, weight $\vec{W}$ acts downward and normal reaction $\vec{N}$ acts upward. Since $\vec{a} = \vec{0}$,

$$\vec{N} + \vec{W} = \vec{0} \quad \text{or in magnitudes} \quad N = W = mg.$$

Correct reasoning

“The book is at rest, therefore $\vec{F}_{\text{net}} = \vec{0}$ and hence $N = mg$ ” is correct. Saying only “forces cancel, therefore the book is at rest” is incomplete because zero net force can also produce uniform non-zero velocity.

4.1 Inertia in a bus

- When a stationary bus starts suddenly, the feet move forward with the bus because of friction, while the upper body tends to remain at rest. The passenger falls backward relative to the bus.
- When a moving bus stops suddenly, the feet stop with the floor, while the upper body tends to continue forward. The passenger falls forward.

Velocity and acceleration are different

At the highest point of a vertical throw, $\vec{v} = \vec{0}$ only for an instant, but $\vec{a} = \vec{g}$ remains vertically downward. Zero velocity does not automatically mean zero force.

5. Momentum

$$\vec{p} = m\vec{v}$$

- Momentum is a **vector quantity**; its direction is the direction of velocity.
- SI unit: kg m s^{-1} . Dimensional formula: $[MLT^{-1}]$.
- A loaded truck is harder to start or stop than a small car moving at the same speed because its momentum is greater.
- For the same mass, a faster body has greater momentum.
- Momentum changes when its magnitude or direction changes. In uniform circular motion, speed is constant but momentum changes continuously because direction changes.

6. Newton's Second Law of Motion

Statement

The rate of change of momentum of a body is directly proportional to the applied net force and takes place in the direction of that force.

6.1 Derivation for constant mass

If momentum changes by $\Delta\vec{p}$ during a small interval Δt , then

$$\vec{F} \propto \frac{\Delta\vec{p}}{\Delta t},$$

$$\vec{F} = k \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{p}}{\Delta t} = k \frac{d\vec{p}}{dt}.$$

In SI units, $k = 1$. Therefore,

$$\vec{F} = \frac{d\vec{p}}{dt}$$

For constant mass, $\vec{p} = m\vec{v}$:

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m \frac{d\vec{v}}{dt} = m\vec{a}.$$

$$\vec{F} = m\vec{a}$$

$$1 \text{ N} = 1 \text{ kg m s}^{-2}.$$

6.2 Important features

- $\vec{F}$ means the **vector sum of all external forces** on the chosen body or system.
- It is a vector law:

$$F_x = ma_x, \quad F_y = ma_y, \quad F_z = ma_z.$$

- Acceleration is parallel to net force, but force need not be parallel to velocity.
- If $\vec{F} = \vec{0}$, then $\vec{a} = \vec{0}$, so the second law agrees with the first law.
- For a system of particles, use total external force and acceleration of the system as a whole. Internal forces are excluded.
- The law is local: the force acting here and now determines acceleration here and now. A body does not carry a force from its past motion.

6.3 Force not parallel to velocity

- In projectile motion without air resistance, gravity is vertical, so the horizontal component of velocity remains unchanged.
- In uniform circular motion, force is perpendicular to velocity. It changes direction without changing speed.

7. Impulse

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt = \Delta\vec{p}$$

For average force:

$$\vec{J} = \vec{F}_{\text{avg}} \Delta t = \vec{p}_f - \vec{p}_i$$

- An impulsive force is large, acts for a short time and produces a finite change in momentum.
- Examples include a bat hitting a ball, a ball bouncing from a wall and a hammer striking a nail.
- SI unit: $\text{N s} = \text{kg m s}^{-1}$. Impulse is a vector.
- A cricketer moves the hands backward while catching. The same $\Delta\vec{p}$ occurs over a larger Δt , so F_{avg} is smaller.
- If a ball reverses direction, the initial and final momentum vectors have opposite signs. Use $\Delta\vec{p} = \vec{p}_f - \vec{p}_i$ carefully.

Force-time graph

The signed area under a force-time graph is impulse:

$$\vec{J} = \int \vec{F} dt.$$

A narrow high pulse and a wider low pulse can produce the same impulse if their areas are equal.

8. Newton's Third Law of Motion

Statement

For every force exerted by body A on body B , body B simultaneously exerts an equal and opposite force on body A .

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

- Forces arise from mutual interaction and never occur singly.
- “Action” and “reaction” are only names. Either force may be called the action.
- They act simultaneously. There is no time delay and no cause-effect sequence.
- They act on **different bodies**; therefore they cannot cancel in the equation of motion of one body.
- For the combined system $A + B$, the pair is internal and cancels in the total force.
- The Earth pulls a stone downward and the stone pulls the Earth upward with equal force. The Earth's acceleration is negligible because its mass is enormous.

Test for a third-law pair

Write the first force as “force on A by B ”. Its partner must be “force on B by A ”. The two forces must be of the same type, equal in magnitude, opposite in direction and acting on different bodies.

9. Conservation of Momentum

For two interacting bodies A and B during the same small interval Δt ,

$$\Delta \vec{p}_A = \vec{F}_{AB} \Delta t,$$

$$\Delta \vec{p}_B = \vec{F}_{BA} \Delta t.$$

Using $\vec{F}_{AB} = -\vec{F}_{BA}$,

$$\Delta \vec{p}_A = -\Delta \vec{p}_B.$$

Therefore,

$$\vec{p}'_A - \vec{p}_A = -(\vec{p}'_B - \vec{p}_B),$$

$$\vec{p}_A + \vec{p}_B = \vec{p}'_A + \vec{p}'_B.$$

Law of conservation of momentum

The total momentum of an isolated system remains constant:

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}}$$

The net external force or external impulse must be zero or negligible.

9.1 Applications

- Gun recoil:** if gun and bullet start from rest, their final momenta are equal and opposite:

$$m_b \vec{v}_b + m_g \vec{v}_g = \vec{0}.$$

- Collision:** total momentum before collision equals total momentum after collision, whether the collision is elastic or inelastic.
- Explosion or nuclear disintegration:** if the original body is at rest, the vector sum of momenta of all fragments is zero.
- In an **elastic collision**, kinetic energy is also conserved. Momentum conservation alone applies to both elastic and inelastic collisions when external impulse is negligible.

10. Equilibrium of a Particle

$$\sum \vec{F} = \vec{0}$$

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum F_z = 0.$$

- A particle in equilibrium may be at rest or may move with constant velocity.
- For two forces, equilibrium requires equal magnitudes and opposite directions.

- For three concurrent forces, the resultant of any two must be equal and opposite to the third.
- For many forces, the force vectors form a closed polygon when placed head-to-tail.

Particle and rigid body

For a particle, $\sum \vec{F} = \vec{0}$ is sufficient. For an extended rigid body, rotational equilibrium is also required: $\sum \vec{\tau} = \vec{0}$.

11. Common Forces in Mechanics

Force	Meaning and direction
Weight $\vec{W} = m\vec{g}$	Gravitational force of Earth on a body; vertically downward near Earth.
Normal reaction $\vec{N}$	Contact-force component perpendicular to the surfaces in contact.
Friction $\vec{f}$	Contact-force component parallel to the surfaces; opposes impending or actual relative sliding.
Tension $\vec{T}$	Pull exerted by a stretched string; acts along the string away from the body.
Spring force	Restoring force $\vec{F} = -k\vec{x}$ for small deformation.
Buoyant force	Upward force on a body immersed in a fluid; equal to the weight of displaced fluid.
Viscous drag	Fluid contact force opposing relative motion through the fluid.
Centripetal force	Name for the net inward force required for circular motion; not a new fundamental force.

- Gravity acts at a distance and does not require an intervening medium.
- Normal reaction, friction, elasticity and impact arise microscopically mainly from electrical interactions among charged constituents of matter.
- For a light, inextensible string passing over a smooth pulley, tension is treated as the same throughout the string.

12. Friction

12.1 Static friction

- Static friction acts when surfaces do not slide relative to each other.
- It opposes **impending relative motion**: the motion that would occur if friction were absent.
- It is self-adjusting and increases only as much as needed, up to a maximum.

$$0 \leq f_s \leq f_{s,\max} = \mu_s N$$

12.2 Kinetic or sliding friction

- Once sliding begins, kinetic friction opposes actual relative motion.
- It is approximately independent of contact area and nearly independent of ordinary sliding speed.

$$f_k = \mu_k N, \quad \mu_k < \mu_s$$

Friction opposes relative motion

A box resting on the floor of an accelerating train is accelerated forward by static friction. Friction is not always opposite to the motion of the object; it is opposite to the tendency of relative slipping between surfaces.

12.3 Angle of repose

For a block on an inclined plane,

$$mg \sin \theta \text{ acts down the plane,} \quad N = mg \cos \theta.$$

At the instant the block is about to slide,

$$\begin{aligned} mg \sin \theta_{\max} &= f_{s,\max} = \mu_s N, \\ mg \sin \theta_{\max} &= \mu_s mg \cos \theta_{\max}. \end{aligned}$$

Hence,

$$\tan \theta_{\max} = \mu_s.$$

The limiting angle depends on the surfaces, not on the mass of the block.

12.4 Maximum acceleration of a train

If a box does not slip on the floor of an accelerating train, static friction supplies its acceleration:

$$ma = f_s \leq \mu_s N = \mu_s mg.$$

Therefore,

$$a_{\max} = \mu_s g.$$

12.5 Rolling friction

- Real rolling surfaces deform slightly, producing rolling resistance.
- Rolling friction is much smaller than static or sliding friction, often by two or three orders of magnitude.
- Wheels and ball bearings reduce resistance by replacing sliding with rolling.

12.6 Useful and harmful effects

Useful friction	Harmful friction and reduction
Walking, tyre grip, vehicle acceleration and brakes	Heat, wear and power loss in machines
Holding objects and writing	Lubricants reduce sliding friction
Holding a body at rest on an inclined plane	Ball bearings reduce resistance
Providing centripetal force on a level road	An air cushion separates moving surfaces

13. Circular Motion and Centripetal Force

For uniform circular motion of radius R and speed v ,

$$a_c = \frac{v^2}{R}, \quad F_c = ma_c = \frac{mv^2}{R}.$$

Both $\vec{a}_c$ and the net centripetal force point towards the centre.

Centripetal force is not an extra force

- Tension supplies it for a stone on a string.
- Gravity supplies it for a planet.
- Friction supplies it for a car on a level road.

Write “net inward force = mv^2/R ”, then identify the real forces that contribute.

13.1 Car on a level circular road

The vertical forces balance:

$$N - mg = 0 \quad \Rightarrow \quad N = mg.$$

Static friction supplies the inward force:

$$\frac{mv^2}{R} = f_s \leq \mu_s N = \mu_s mg.$$

Therefore,

$$v \leq v_{\max} = \sqrt{\mu_s Rg}.$$

- The limiting speed is independent of the car's mass.

- Smaller R means a sharper turn and a lower safe speed.
- Smaller μ_s on a wet or slippery road also lowers the safe speed.

14. Car on a Banked Road

A road is banked by raising the outer edge. The normal reaction then has a horizontal component towards the centre, reducing dependence on friction.

At the maximum-speed condition, friction acts down the slope and has limiting magnitude $f = \mu_s N$:

$$\begin{aligned} N \cos \theta &= mg + f \sin \theta, \\ N \sin \theta + f \cos \theta &= \frac{mv^2}{R}. \end{aligned}$$

Substituting $f = \mu_s N$ gives

$$v_{\max} = \sqrt{Rg \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}.$$

14.1 Optimum speed without friction

For $f = 0$,

$$N \cos \theta = mg, \quad N \sin \theta = \frac{mv_0^2}{R}.$$

Dividing the equations,

$$\tan \theta = \frac{v_0^2}{Rg} \quad \Rightarrow \quad v_0 = \sqrt{Rg \tan \theta}.$$

- At v_0 , normal reaction alone provides the centripetal force and sideways tyre wear is minimal.
- For $v < v_0$, friction acts up the slope.
- A car can remain parked only if $\tan \theta \leq \mu_s$.

15. Free-Body Diagrams and Problem Solving

Free-body diagram (FBD)

An FBD isolates one chosen body or system and shows every external force acting on it. It does not show forces exerted by that system on the environment. "Free-body" does not mean force-free.

15.1 Systematic method

1. Draw the physical arrangement: bodies, strings, pulleys, surfaces, supports and expected motion.
2. Choose the body or group of bodies that will be your system.
3. Draw its FBD with weight, normal reaction, tension, friction and other external forces.
4. Choose convenient axes, often along and perpendicular to the motion or inclined plane.
5. Resolve angled forces into components.
6. Apply

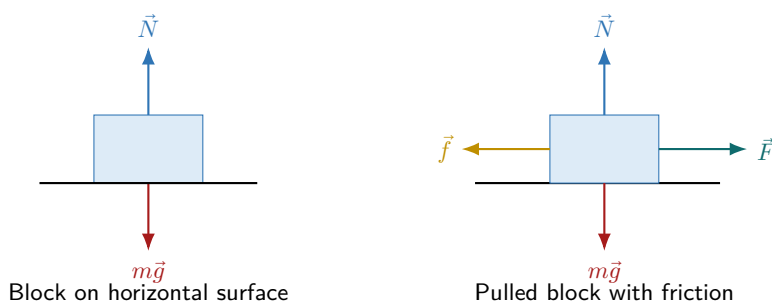
$$\sum F_x = ma_x, \text{quad} \sum F_y = ma_y.$$

7. Use constraints: connected bodies may have the same acceleration magnitude; a light inextensible string over a smooth pulley has a common tension.
8. If necessary, draw another FBD for a second body and connect the equations.
9. Check units, signs, directions, limiting cases and physical sense.

Do not place these on an FBD

- $m\vec{a}$ is not an extra force. It is the result of the net force.
- Do not place both members of a third-law pair on one body.
- Do not include internal forces when the chosen system already contains both interacting bodies.

15.2 Basic force-diagram patterns



16. Standard Numerical Patterns

16.1 Stopping under a constant retarding force

Choose positive direction along the initial velocity. Use

$$v^2 = u^2 + 2as \quad \text{or} \quad v = u + at,$$

then find $\vec{F} = m\vec{a}$. A negative answer means the force is opposite to the chosen positive direction.

16.2 Impulse during rebound

Choose a sign convention and calculate

$$\vec{J} = m(\vec{v} - \vec{u}).$$

When direction reverses, the two velocities have opposite signs.

16.3 Connected bodies

- Treat the entire system first if this removes internal tension.
- Find acceleration from the net external force.
- Then isolate one body to calculate the tension.

16.4 Block on an inclined plane

$$W_{\parallel} = mg \sin \theta, \quad W_{\perp} = mg \cos \theta.$$

If no other perpendicular force acts, $N = mg \cos \theta$. Decide the tendency of slipping before assigning the direction of friction.

16.5 Lift and weighing-machine problems

The reading is the normal reaction N , not automatically mg :

Upward acceleration:	$N = m(g + a),$
Downward acceleration:	$N = m(g - a),$
Uniform speed:	$N = mg,$
Free fall:	$N = 0.$

16.6 Circular-turn problems

Convert speed to m s^{-1} , then write

$$\sum F_{\text{inward}} = \frac{mv^2}{R}.$$

On a level road, limiting static friction gives maximum speed. On a friction-free banked road, use $v_0 = \sqrt{Rg \tan \theta}$.

17. Formula Sheet with Conditions

Quantity / case	Formula	Condition or meaning
Momentum	$\vec{p} = m\vec{v}$	Vector definition for a particle.

Second law	$\vec{F} = \frac{d\vec{p}}{dt}$	$\vec{F}$ is the net external force.
Constant mass	$\vec{F} = m\vec{a}$	Used when mass is constant.
Impulse	$\vec{J} = \int \vec{F} dt = \Delta\vec{p}$	Vector; force may vary during contact.
Momentum conservation	$\sum \vec{p}_i = \sum \vec{p}_f$	External impulse is zero or negligible.
Particle equilibrium	$\sum \vec{F} = \vec{0}$	Rest or uniform velocity.
Spring force	$\vec{F} = -k\vec{x}$	Small elastic deformation.
Static friction	$0 \leq f_s \leq \mu_s N$	Self-adjusting; equality only at the limit.
Kinetic friction	$f_k = \mu_k N$	Surfaces are sliding.
Angle of repose	$\tan \theta_{\max} = \mu_s$	Block is just about to slide.
Centripetal force	$F_c = mv^2/R$	Net inward force, not a separate interaction.
Level-road limit	$v_{\max} = \sqrt{\mu_s Rg}$	Static friction provides the inward force.
Banked-road optimum	$v_0 = \sqrt{Rg \tan \theta}$	No friction is required.
Banked-road maximum	$v_{\max} = \sqrt{Rg \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$	Friction acts down the bank at maximum speed.

18. Units and Dimensions

Quantity	SI unit	Dimensions
Momentum	kg m s^{-1}	$[MLT^{-1}]$
Force, weight, tension, normal reaction	newton, N	$[MLT^{-2}]$
Impulse	N s	$[MLT^{-1}]$
Coefficient of friction	no unit	dimensionless
Spring constant k	N m^{-1}	$[MT^{-2}]$
Acceleration	m s^{-2}	$[LT^{-2}]$

19. Most Important Misconceptions

Incorrect idea	Correct understanding
A moving body must have force in the direction of motion.	Force is parallel to acceleration, not necessarily velocity. It may be forward, backward, sideways or zero.
If velocity is zero, force is zero.	At a turning point, velocity may be zero while force and acceleration are non-zero.
A body carries force from its earlier motion.	Force is determined by present interactions at the body's present location.
$m\vec{a}$ is another force on the body.	$m\vec{a}$ equals the net external force; it must not be drawn as a separate force.
Action and reaction cancel on one body.	They act on different bodies. They cancel only as internal forces of the combined system.
Weight and normal reaction are always equal.	They are equal only in appropriate zero-vertical-acceleration situations.
Static friction always equals $\mu_s N$.	It adjusts from zero to the maximum $\mu_s N$.
Friction always opposes motion.	It opposes relative slipping and can accelerate an object.
Centripetal force is a new kind of force.	It is the name of the net inward force supplied by real interactions.
Feeling pushed outward proves an outward force acts.	In an inertial frame the actual net force is inward; the outward feeling is related to inertia.

20. Exam Answer Frameworks

20.1 Why was Aristotle's law wrong?

- State Aristotle's belief that continuous force is needed for continuous motion.
- Explain that ordinary objects stop because of friction and drag.
- During constant velocity, the applied force balances friction, so $\vec{F}_{\text{net}} = \vec{0}$.
- Use Galileo's frictionless-plane reasoning.

- Conclude that force changes velocity; it is not required for unchanged velocity.

20.2 Derive Newton's second law

- Define $\vec{p} = m\vec{v}$.
- State $\vec{F} \propto d\vec{p}/dt$.
- Choose the SI proportionality constant as 1.
- For constant mass, show $d\vec{p}/dt = m d\vec{v}/dt = m\vec{a}$.
- Conclude $\vec{F} = m\vec{a}$ and define 1 N.

20.3 Prove conservation of momentum

- Take two interacting bodies in an isolated system.
- Apply $\vec{F}_{AB} = -\vec{F}_{BA}$.
- Multiply by the common interaction time to obtain $\Delta\vec{p}_A = -\Delta\vec{p}_B$.
- Rearrange to show total initial momentum equals total final momentum.
- State that the net external impulse must be zero or negligible.

20.4 Static, limiting and kinetic friction

- Static friction is self-adjusting: $0 \leq f_s \leq \mu_s N$.
- Limiting friction is the maximum static value: $f_{s,\max} = \mu_s N$.
- Kinetic friction acts during sliding: $f_k = \mu_k N$.
- Usually $\mu_k < \mu_s$, so maintaining sliding is easier than starting it.

20.5 Explain banking of roads

- The outer edge is raised so the normal reaction tilts towards the centre.
- Its horizontal component helps provide centripetal force.
- With no friction, divide $N \sin \theta = mv_0^2/R$ by $N \cos \theta = mg$.
- Obtain $v_0 = \sqrt{Rg \tan \theta}$.
- Banking reduces sideways friction, tyre wear and the chance of skidding.

21. Rapid Revision Sheet

Twelve facts to remember

- Inertia is resistance to change in rest or uniform straight-line motion.
- $\sum \vec{F} = \vec{0}$ means $\vec{a} = \vec{0}$, not necessarily $\vec{v} = \vec{0}$.
- Momentum is $\vec{p} = m\vec{v}$ and is a vector.
- Net force is $d\vec{p}/dt$.
- Impulse is $\Delta\vec{p}$.
- Third-law forces are simultaneous and act on different bodies.
- Momentum is conserved only when external impulse is zero or negligible.
- Equilibrium requires $\sum \vec{F} = \vec{0}$.
- Static friction adjusts up to $\mu_s N$; kinetic friction is $\mu_k N$.
- Friction opposes relative slipping and may accelerate a body.
- Centripetal force is the net inward force, not a separate interaction.
- Every numerical solution should begin by choosing a system and drawing its FBD.

Final exam checklist

- Convert all data to SI units and state the sign convention.
- Identify the body or system clearly and draw all external forces.
- Do not add a separate force called $m\vec{a}$.
- Use the static-friction inequality correctly.
- State magnitude, direction and unit in the final answer.
- Check dimensions and physical sense.